

relatorio

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Estimadores de Mínimos Quadrados

seja X a matriz de dados com n observações e p variáveis;

seja y a variável alvo

Os estimadores de mínimos quadrados são:

$$\hat{B} = (X'X)^{-1} X'y$$

Assumindo o modelo linear $Y = XB + e$,

e assumindo que, para todo i e j :

$$E(e_i) = 0$$

$$\text{cov}(e_i, e_j) = 0 \text{ (erros não correlacionados)}$$

$$\text{var}(e_i) = \sigma^2 \text{ (homocedasticidade)}$$

de acordo com o teorema de Gauss-Markov, os estimadores de mínimos quadrados serão os melhores estimadores lineares não viesados, sob as suposições anteriores.

Para os testes de hipóteses, assumimos que $e_i \sim N(0, \sigma^2)$

Imports

```
#install.packages("plotly")  
library("plotly")
```

```
source("functions.R")
```

Função Gaussiana

```
$$  
f(x, y) = K \cdot e^{-c(x^2 + y^2)}  
$$
```

$$f(x, y) = k \cdot \exp[-c(x^2 + y^2)]; k, c > 0$$

k : Amplitude máxima da função; $f(0,0) = k$

c : Abertura e decaimento da curva;

$f: \mathbb{R}^2 \rightarrow \mathbb{R}^+$

```
f <- function(x,y, k=10, c=6){
  return( k*exp(-c(x^2 + y^2)))
}
```

Objetivo

Usar polinômios de duas variáveis para aproximar a função gaussiana;

Geração de pontos aleatórios

```
set.seed(28)

# número de pontos
n<- 100

min_<- -4
max_<- 4

x_unif<- runif(n, min= min_, max= max_)
y_unif<- runif(n, min= min_, max= max_)
z_unif<- f(x_unif,y_unif)

x_normal<- rnorm(n, 0, 2)
y_normal<- rnorm(n, 0, 2)
z_normal<- f(x_normal, y_normal)

pontos_unif<- cbind(x_unif,y_unif, z_unif)
pontos_normal<- cbind(x_normal, y_normal, z_normal)

matriz_unif<- data.frame(pontos_unif)
matriz_normal<- data.frame(pontos_normal)

print(matriz_unif[1:5,])

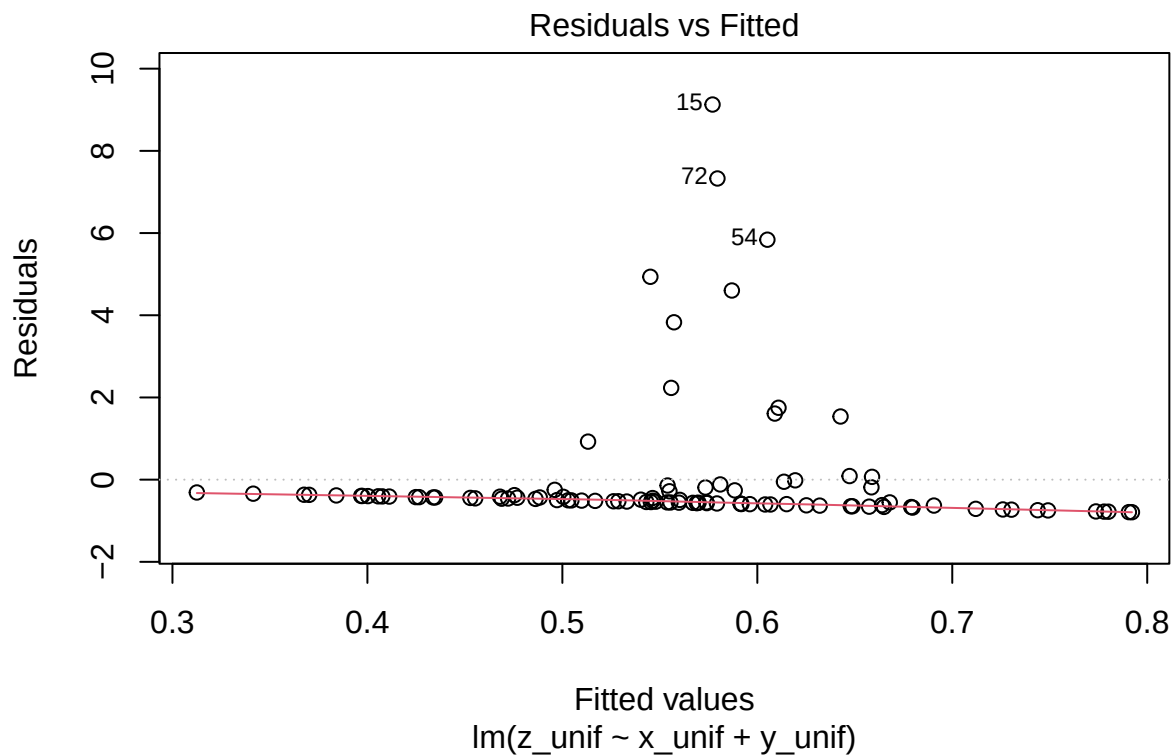
##      x_unif    y_unif      z_unif
## 1 -3.771398  1.964150 1.404043e-07
## 2 -3.293045 -1.324710 3.375378e-05
## 3 -0.205058 -2.320177 4.403878e-02
## 4  3.083021  3.817274 3.496746e-10
## 5 -3.267464  1.095955 6.945364e-05
```

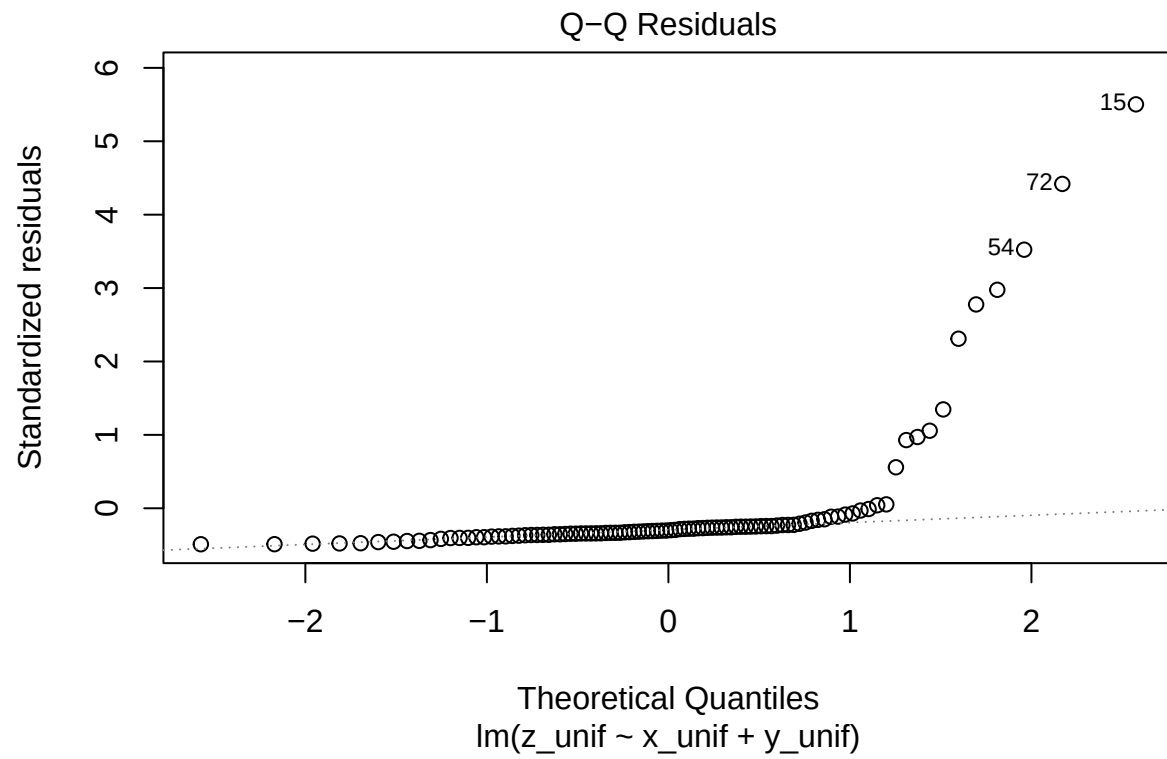
Modelo 1 [$m_1(x,y) = b_0 + b_1x + b_2y$]

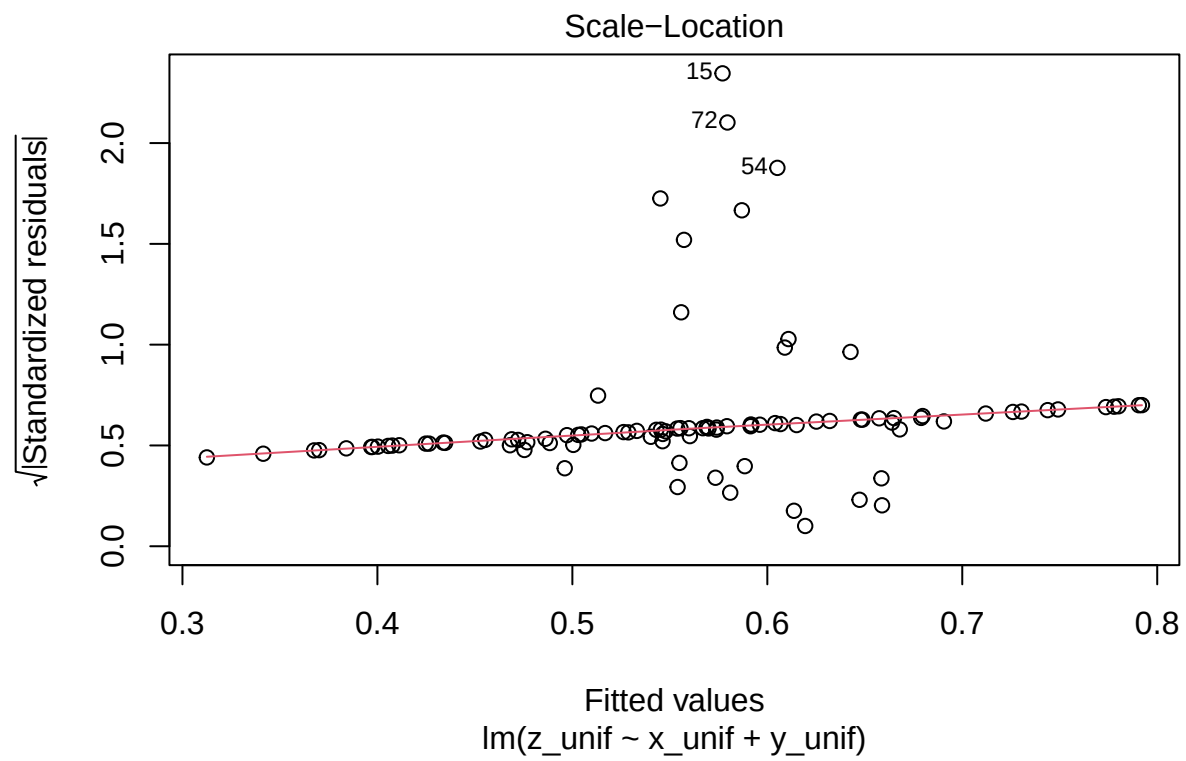
```
modelo1_unif<- lm(z_unif ~ x_unif + y_unif, data=matriz_unif)
modelo1_noraml<- lm(z_normal ~ x_normal + y_normal, data=matriz_normal)
summary(modelo1_unif)

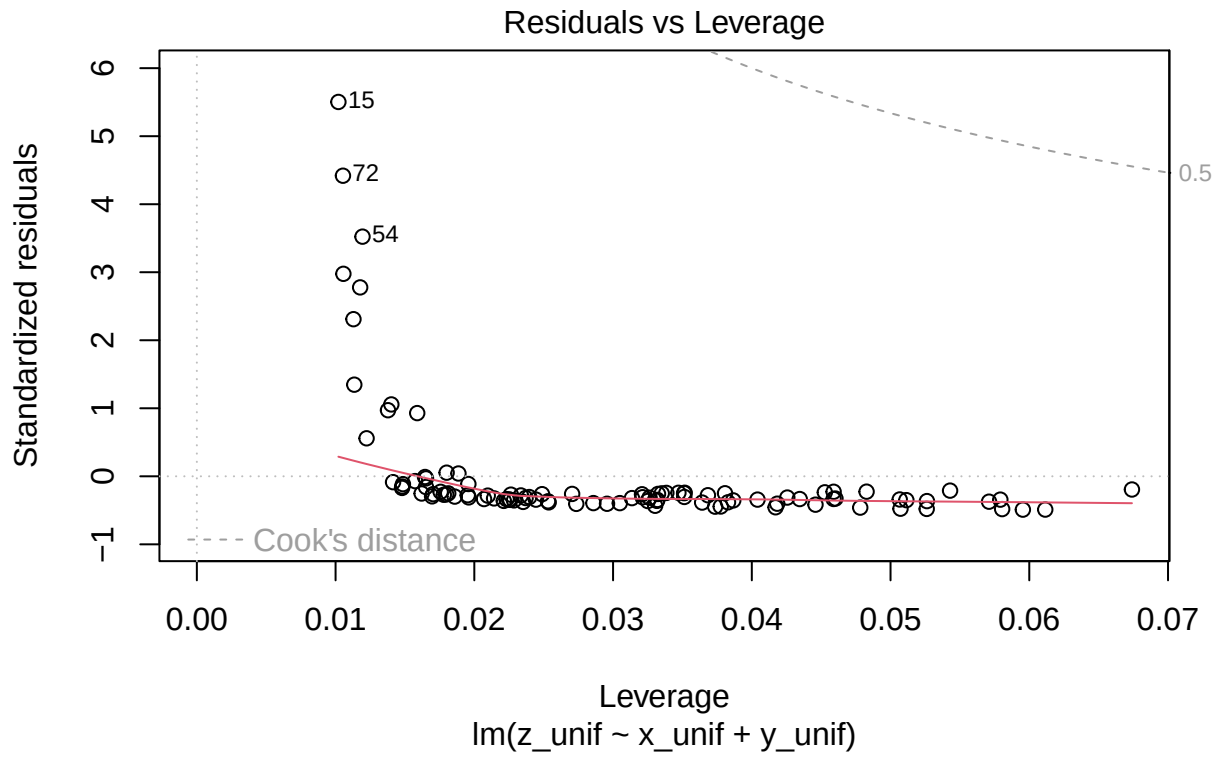
##
## Call:
## lm(formula = z_unif ~ x_unif + y_unif, data = matriz_unif)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.7922 -0.5916 -0.4951 -0.3695  9.1272
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.58110    0.16942   3.43 0.000888 ***
## x_unif         0.03216    0.07472   0.43 0.667805
## y_unif        -0.03856    0.07274  -0.53 0.597208
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.667 on 97 degrees of freedom
## Multiple R-squared:  0.004074,    Adjusted R-squared:  -0.01646
## F-statistic: 0.1984 on 2 and 97 DF,  p-value: 0.8204
plot(modelol_unif)
```









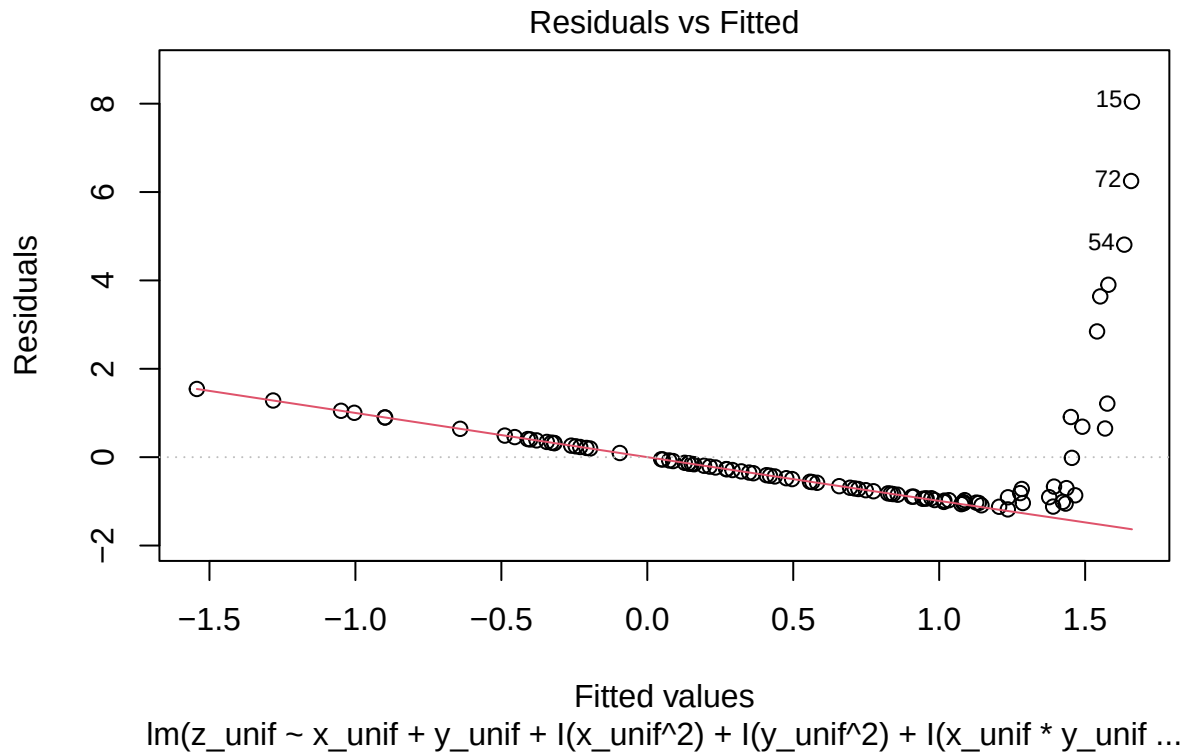
Modelo 2 [$m_2(x,y) = b_0 + b_1x + b_2y + b_3x^2 + b_4y^2 + b_5x*y$]

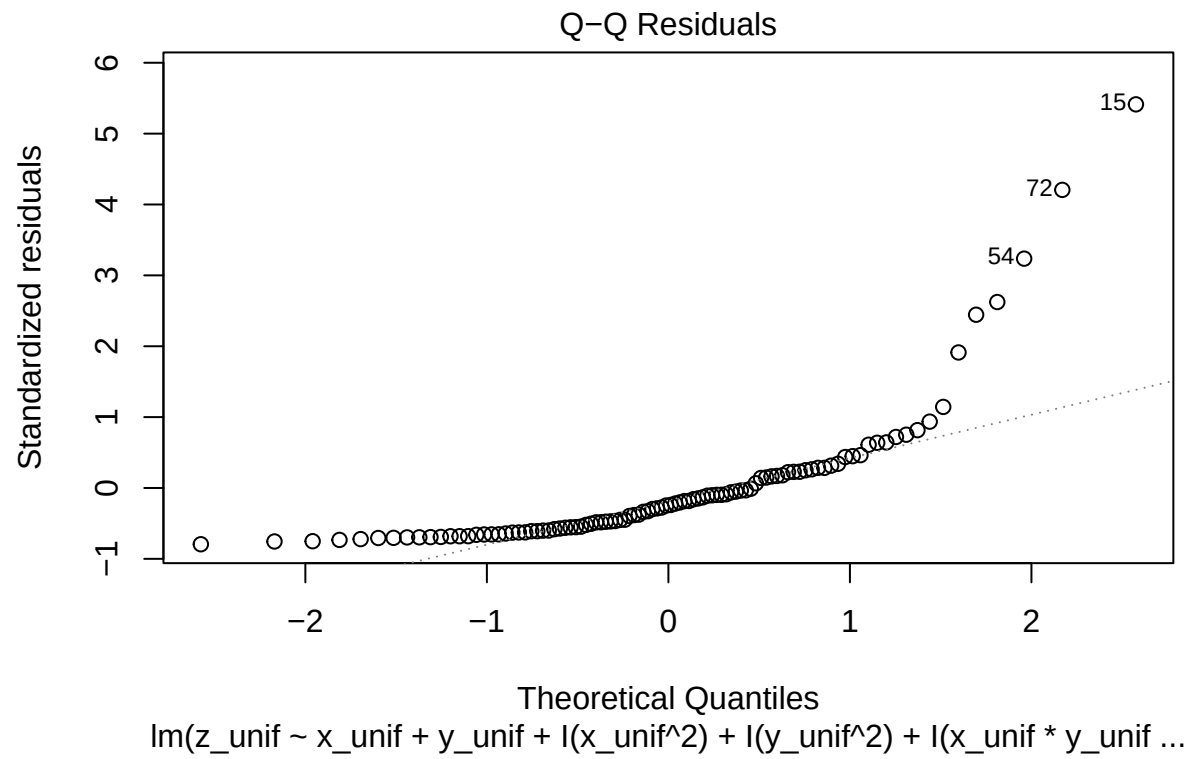
```
modelo2<- lm(z_unif ~ x_unif + y_unif + I(x_unif^2) + I(y_unif^2) + I(x_unif*y_unif), data=matriz_unif)
summary(modelo2)
```

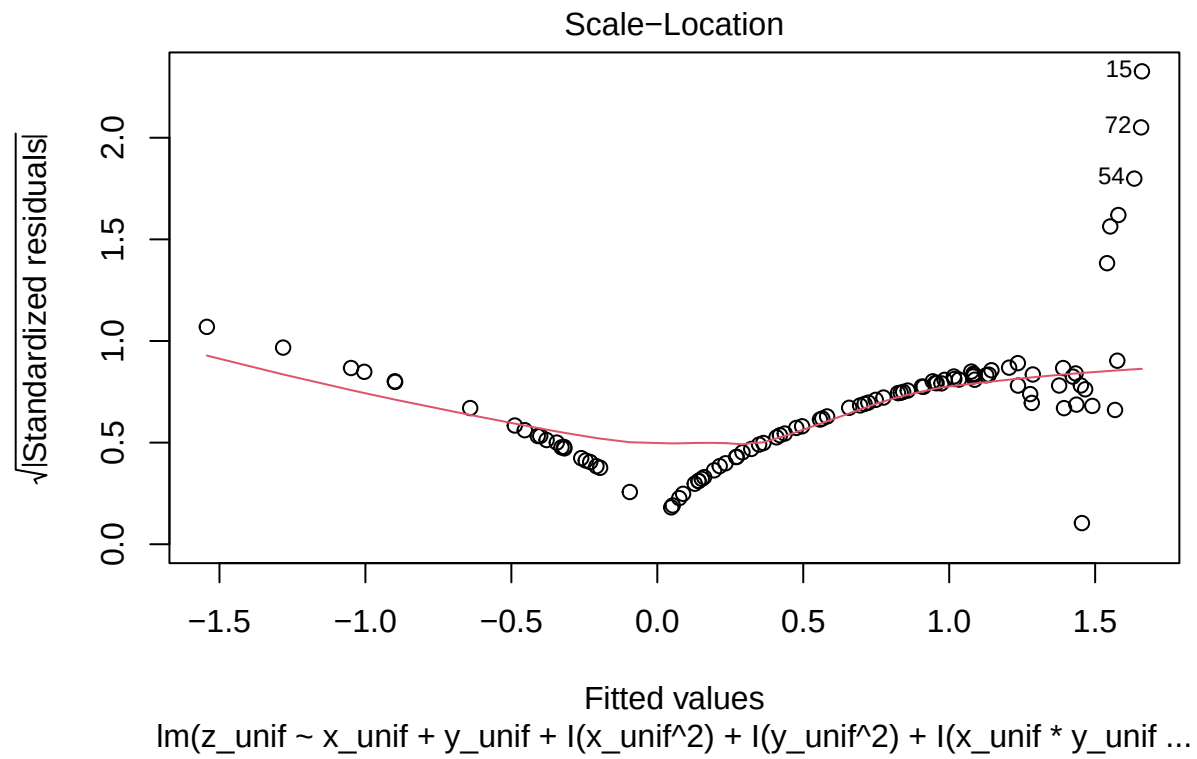
```
##
## Call:
## lm(formula = z_unif ~ x_unif + y_unif + I(x_unif^2) + I(y_unif^2) +
##     I(x_unif * y_unif), data = matriz_unif)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.1823 -0.8937 -0.3560  0.3185  8.0439
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.65558    0.26678   6.206 1.46e-08 ***
## x_unif          0.03650    0.06990   0.522 0.602713
## y_unif          0.03422    0.06789   0.504 0.615417
## I(x_unif^2)    -0.09580    0.03255  -2.943 0.004094 **
## I(y_unif^2)    -0.11118    0.03266  -3.404 0.000979 ***
## I(x_unif * y_unif) 0.01511    0.02811   0.538 0.592143
## ---
```

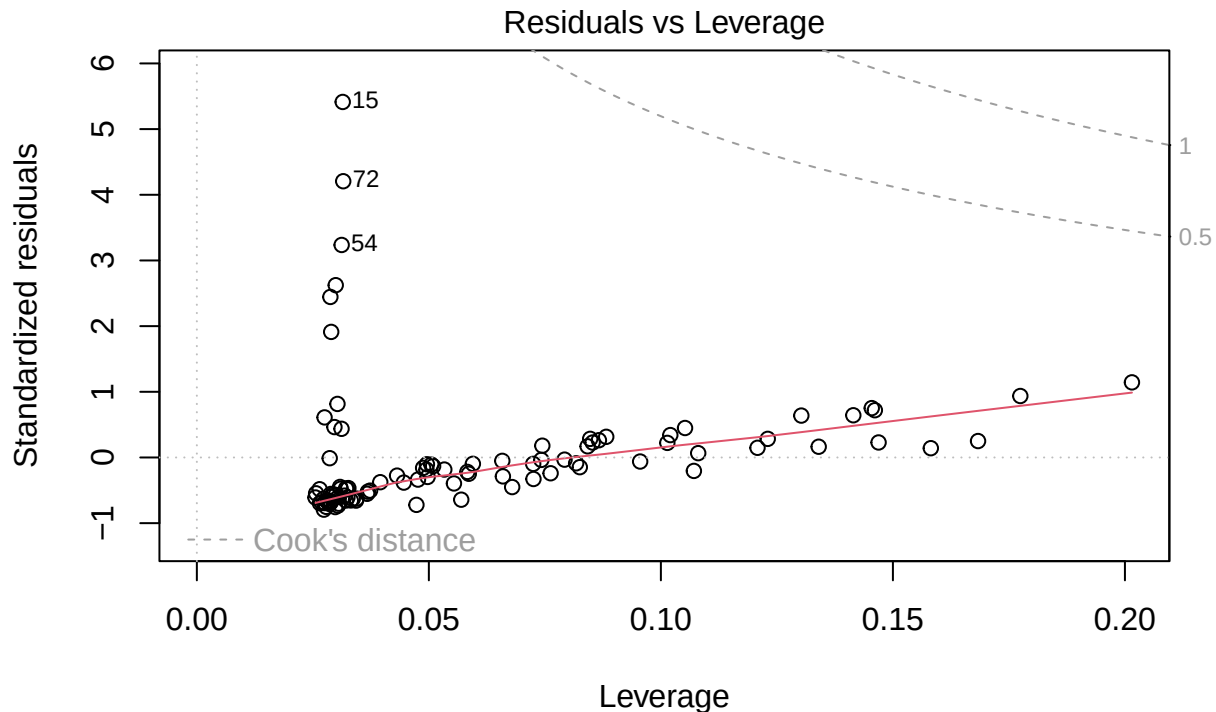
```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.51 on 94 degrees of freedom
## Multiple R-squared:  0.2083, Adjusted R-squared:  0.1662
## F-statistic: 4.947 on 5 and 94 DF,  p-value: 0.0004591
```

```
plot(modelo2)
```









lm(z_unif ~ x_unif + y_unif + I(x_unif^2) + I(y_unif^2) + I(x_unif * y_unif ...

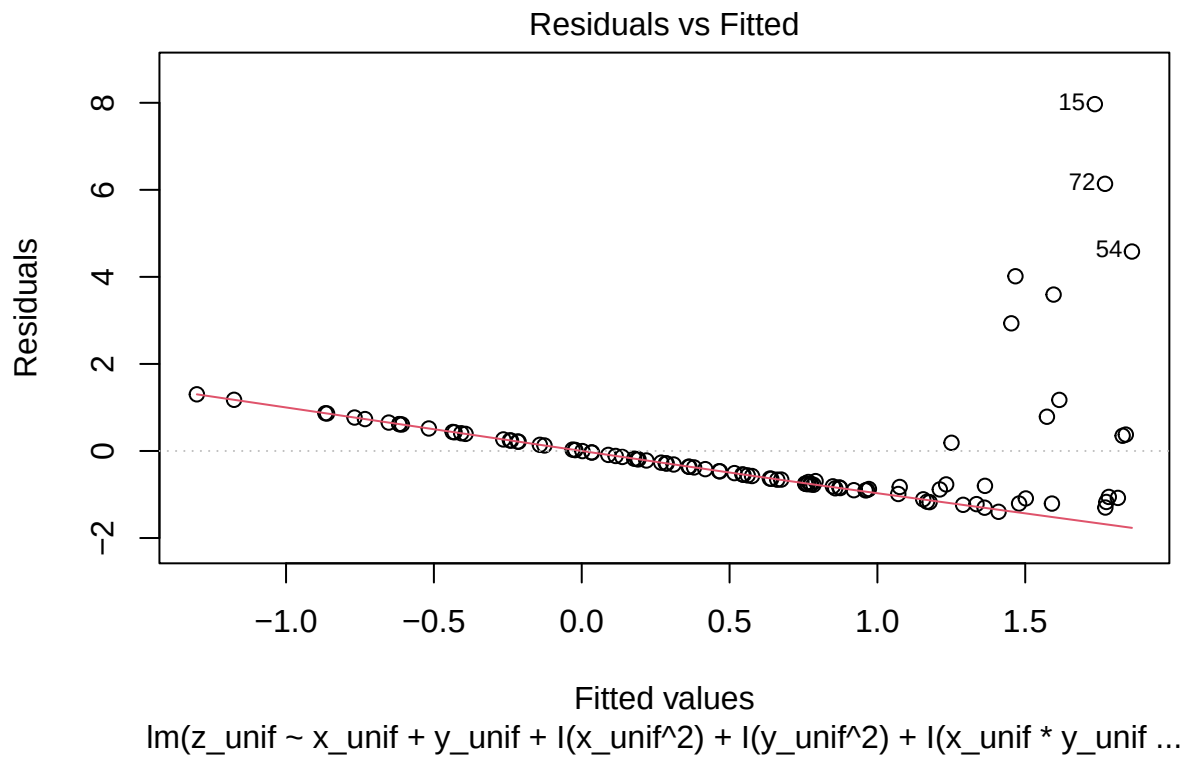
Modelo 3 [$m_2(x,y) = b_0 + b_1 \cdot x + b_2 \cdot y + b_3 \cdot x^2 + b_4 \cdot y^2 + b_5 \cdot x \cdot y + b_6 \cdot x^3 + b_7 \cdot y^3$]

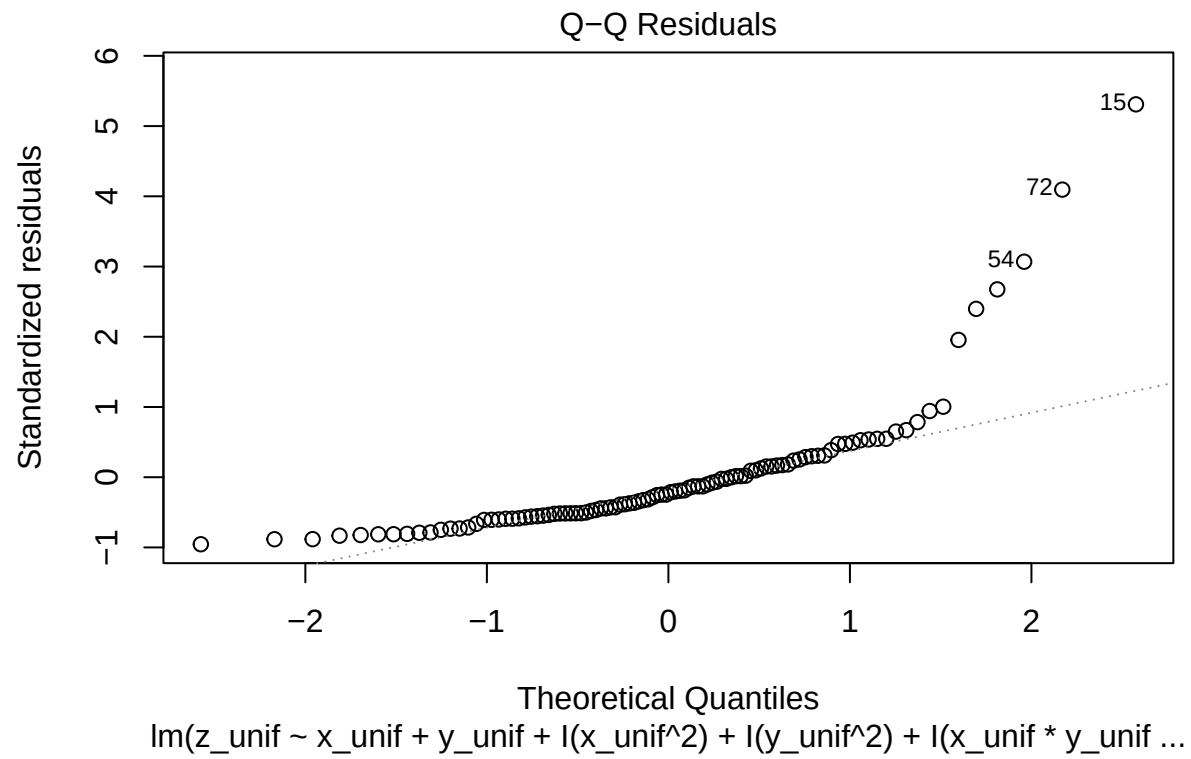
```
modelo3<- lm(z_unif ~ x_unif + y_unif + I(x_unif^2) + I(y_unif^2) + I(x_unif*y_unif) + I(x_unif^3) + I(y_unif^3) + I(x_unif^2 * y_unif) + I(x_unif * y_unif^2), data = matriz_unif)
summary(modelo3)
```

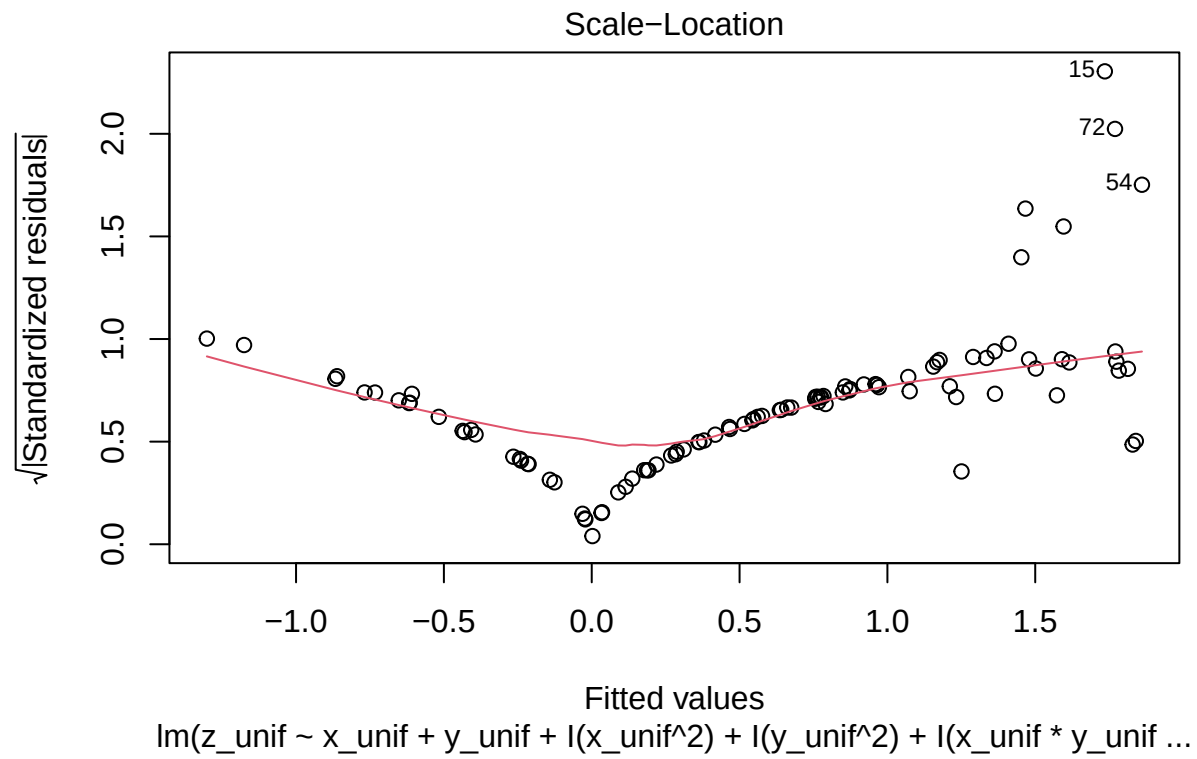
```
##
## Call:
## lm(formula = z_unif ~ x_unif + y_unif + I(x_unif^2) + I(y_unif^2) +
##      I(x_unif * y_unif) + I(x_unif^3) + I(y_unif^3) + I(x_unif^2 *
##      y_unif) + I(x_unif * y_unif^2), data = matriz_unif)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3973 -0.8040 -0.3362  0.2862  7.9689
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.73804    0.27603   6.297 1.09e-08 ***
## x_unif         0.25301    0.18261   1.386 0.169316
## y_unif        -0.09988    0.17544  -0.569 0.570545
## I(x_unif^2)    -0.10230    0.03341  -3.062 0.002902 **
## I(y_unif^2)    -0.12175    0.03422  -3.558 0.000598 ***
## I(x_unif * y_unif)  0.02340    0.03022   0.774 0.440929
```

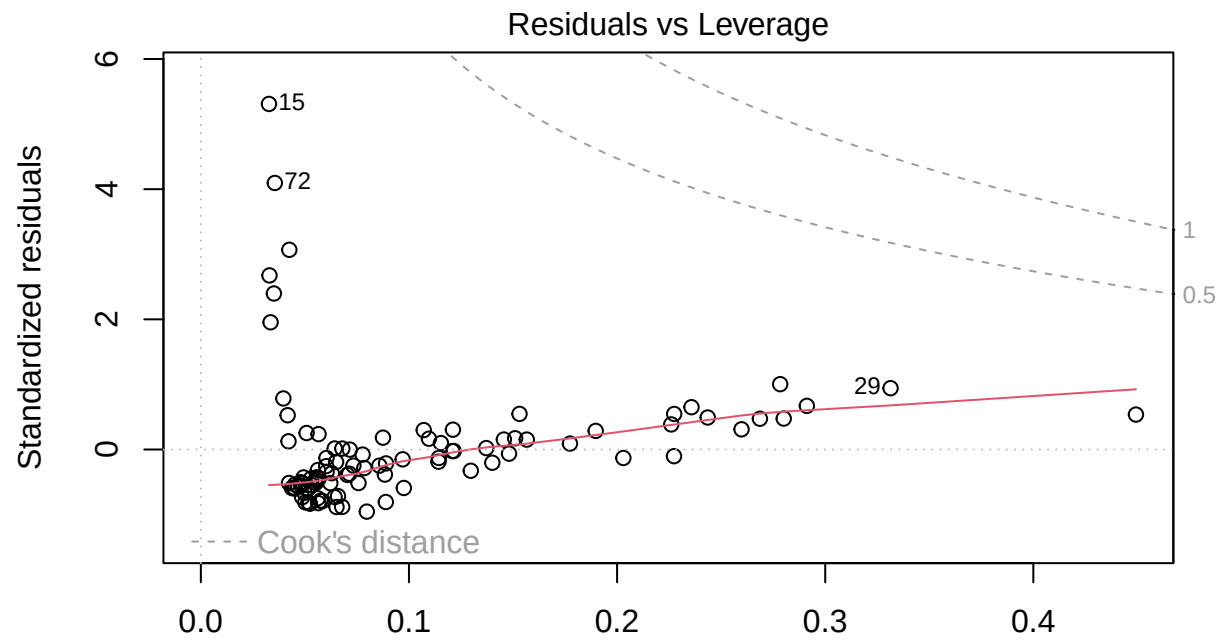
```
## I(x_unif^3)      -0.01638    0.01611   -1.017  0.311967
## I(y_unif^3)      0.01012    0.01564    0.647  0.519286
## I(x_unif^2 * y_unif) 0.00931    0.01404    0.663  0.508863
## I(x_unif * y_unif^2) -0.01164    0.01520   -0.766  0.445810
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.526 on 90 degrees of freedom
## Multiple R-squared:  0.2257, Adjusted R-squared:  0.1483
## F-statistic: 2.915 on 9 and 90 DF,  p-value: 0.004481
```

```
plot(modelo3)
```









$\text{lm}(z_unif \sim x_unif + y_unif + I(x_unif^2) + I(y_unif^2) + I(x_unif * y_unif) \dots)$

```
hist(modelo3$residuals)
```



Observações sobre esses modelos:

$\text{residuo} = \text{valor_real} - \text{valor_previsto}$

$\text{residuo} = -\text{valor_previsto}$ (quando o real é zero)

- A maioria desses modelos não atendem às suposições necessárias para o teorema de gauss-markov
 - Os erros são heterocedásticos e correlacionados.
- O modelo prevê valores menores que zero (fora do contradomínio de f)
- Como a variável Y é truncada assintoticamente em zero, usar mínimos quadrados diretamente não é bom.

Transformação $\text{LN}(Y)$